

INTRODUCTION

To assess the performance of dermal/transdermal formulation, *in vitro* permeation test (IVPT) with Franz cells is a reliable, reproducible, and widely used technique¹. Due to its histological similarity, pig skin is a common model used as an alternative to human skin². However, careful consideration should be given to the pig skin preparation, as both permeability and lag time

can be greatly affected. The main variation sources are pig skin storage time, skin integrity and thickness, sample quantity and methodology to apply the formulation on the membrane³. This study aims to investigate the critical parameters affecting the data's accuracy by evaluating the permeation of a lidocaine hydrochloride-based gel (LID HCl) on pig skin.

MATERIALS AND METHODS

To check the skin integrity, visual assessment, transepidermal water loss (TEWL) with a vapometer (Delphin) and skin's thickness are assessed. The dermal permeation of LID HCl gel (20 µL) was studied on the dermatomed pig skin at 32 ± 0.5°C using PBS buffer pH 7.4 as receptor fluid, constantly stirred at 400 rpm. The IVPT was carried out with vertical Franz diffusion cells set up (n = 6)

with a diffusion surface of 1.76 cm². Samples were withdrawn over a 24h period by an automatic microette Vision system (Synergy) with volume replacement. The drug is quantified by HPLC, coupled to UV/VIS detection. The data were examined statistically using an ANOVA test.

RESULTS

The permeation capacity of a LID HCl based gel was tested over time to check the impact of skin storage conditions on skin permeation property. The pig dorsal skin was cut into 30 strips (displaying the same surface) and kept frozen at -20 °C for several months. For IVPT, the skin strip was washed, dermatomed and punched according to a standardized internal protocol at t0, 1, 3, 4, 5, 6 and 7 months. As shown in Figure 1, the diffusion of LID HCl remains similar, with same lag time, except for the 4 months trial. This performance gap is due to the integrity of the skin used (Figure 2e).

Statistical analysis was then conducted to deeply examine the influence of storage, skin area, quantity of formulation, skin thickness and integrity on LID HCl diffusion. The Q24 value (cumulative amount permeated in 24 hr) was selected as response.

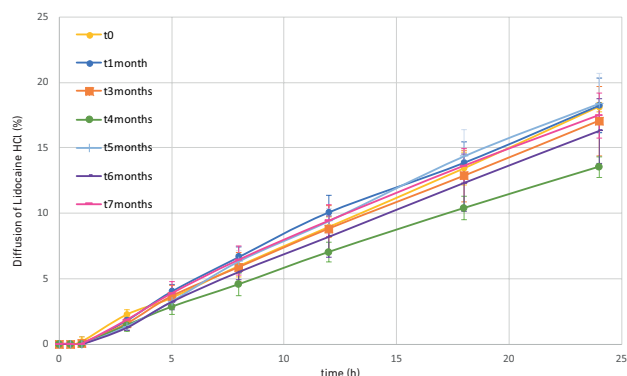


Figure 1. Diffusion of LID HCl based gel regarding pig skin storage time.

Figure 2 shows that storage time (2a), amount of formulation (2b), TEWL (2c) and skin thickness (2d) do not impact diffusion. However, Figure 2e highlights the significant difference in permeability across the pig skin regions. The skin areas located near the head or tail of the pig have different permeation properties. This is certainly due to an alteration during cutting at the slaughterhouse. These areas of the skin are therefore excluded from the study.

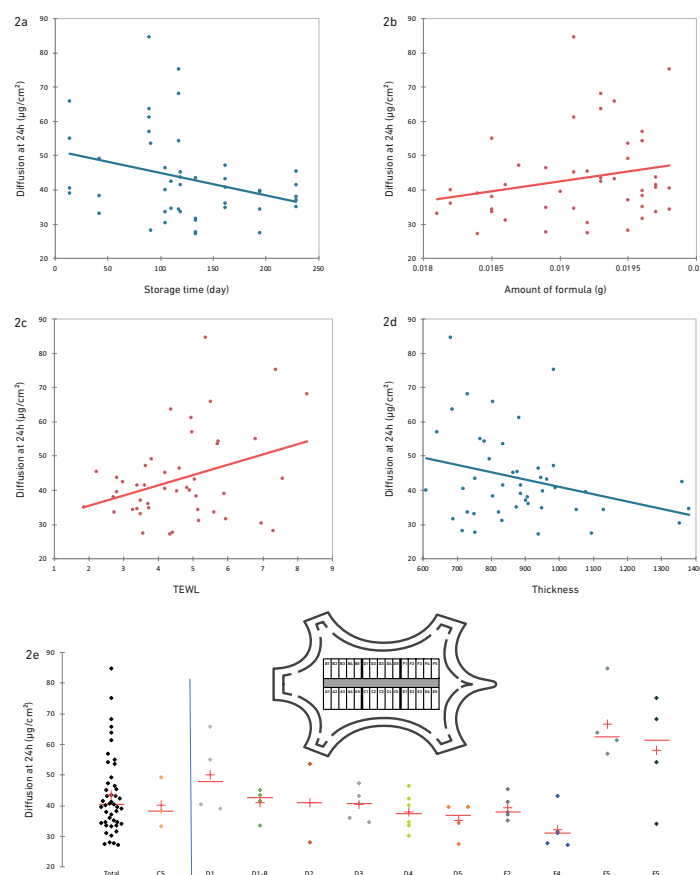


Figure 2. Statistical analysis of different pig skin features on LID HCl permeation.

CONCLUSION

In the conditions used in our laboratory, IVPT can be conducted up to 7 months, following internal specifications (storage, skin area, quantity of formulation, skin thickness and integrity).

To get reliable data, each laboratory should carefully standardize and validate their protocols for *in vitro* permeation studies on animal skin.

REFERENCES

- European Medicines Agency: Draft Guideline on quality and equivalence of topical products EMA/CHMP/QWP/708282/2018
- Neupane, Rabin et al. "Alternatives to Biological Skin in Permeation Studies: Current Trends and Possibilities." *Pharmaceutics* vol. 12, 2 152. 13 Feb. 2020, doi:10.3390/pharmaceutics12020152

- Esposito Biondo, Nicole et al. "How to define the experimental conditions of skin permeation assays for drugs presenting biopharmaceutical limitations? The experience with testosterone." *International journal of pharmaceutics* vol. 607 (2021): 120987. doi:10.1016/j.jipharm.2021.120987